

Design Principles and Usability Analysis of Virtual Reality Manufacturing Digital Twins

Rumena Begum, PhD; Siddhi Kabadi, Faisal Aqlan, PhD; University of Louisville



1 INTRODUCTION

- Virtual Reality (VR) enables immersive and interactive manufacturing environments for training and system monitoring.
- Digital Twins (DTs) provide real-time synchronization between physical and virtual manufacturing systems.
- Poor usability can reduce the effectiveness of VR-based DT systems

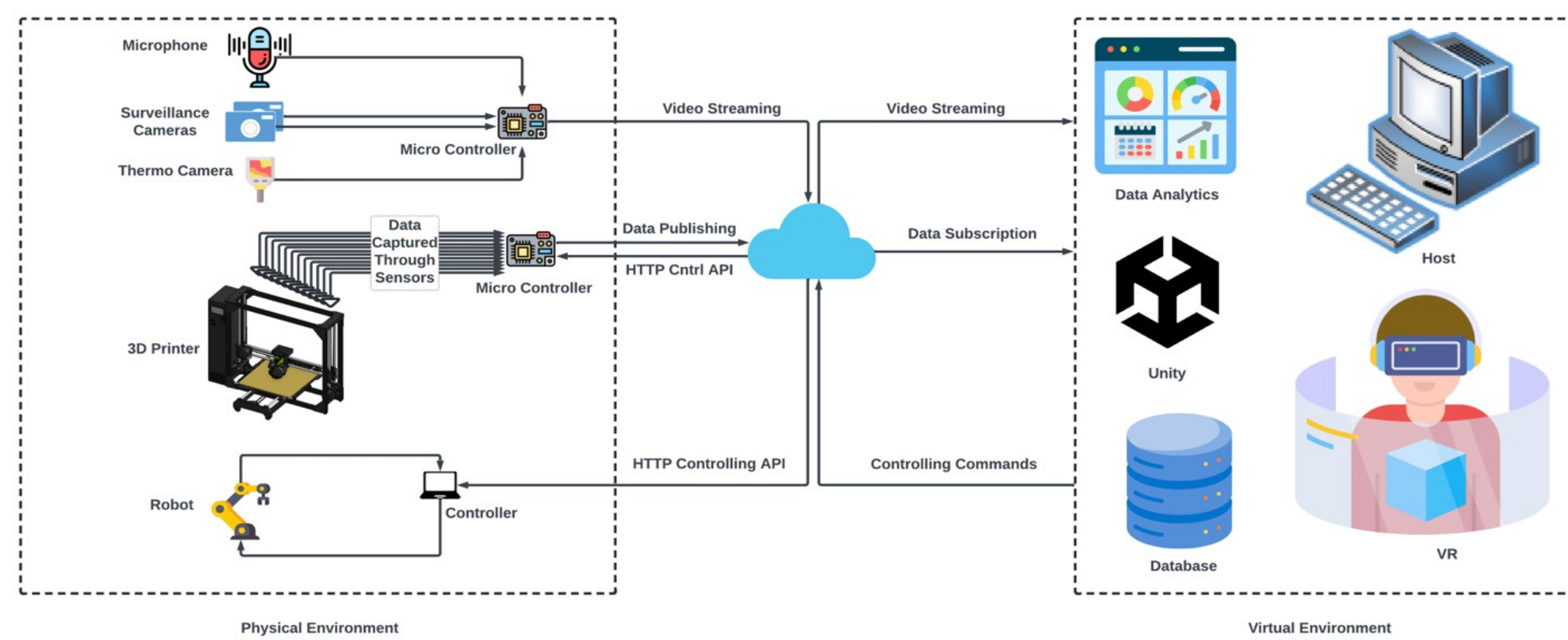
2 PROJECT OBJECTIVES

- Develop a VR-based digital twin of a collaborative manufacturing system with robots and 3D printers
- Evaluate usability, workload, and user interaction during VR manufacturing tasks
- Identify design insights to support more effective and human-centered VRDT systems

3 CHALLENGES

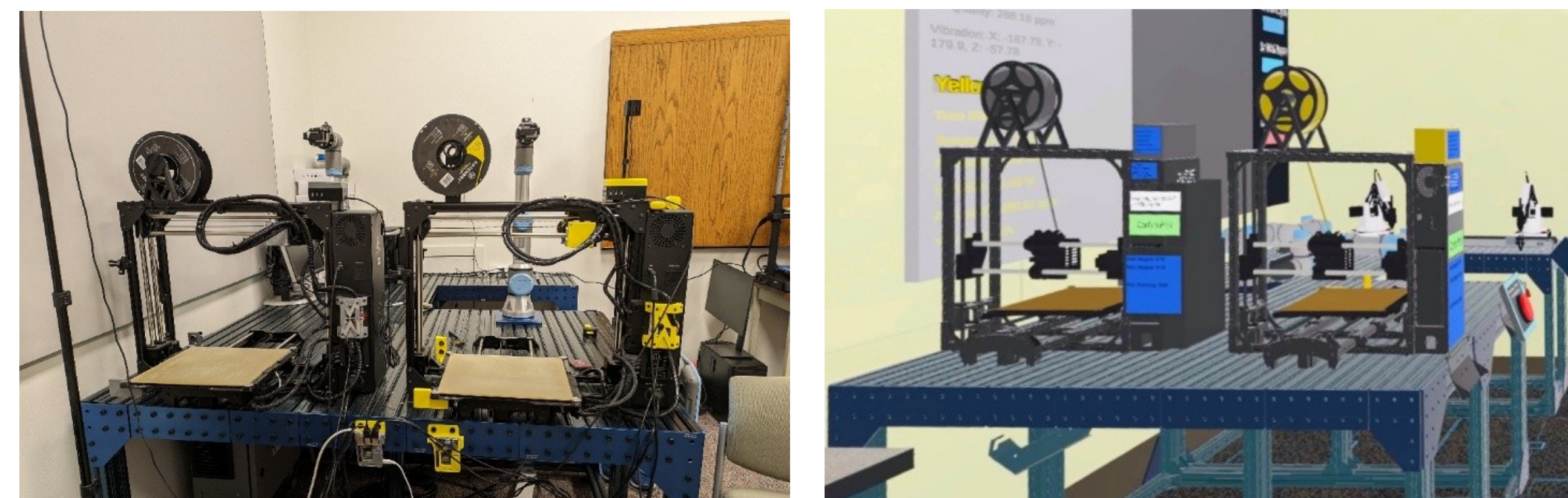
- Real-time synchronization and high-fidelity VR simulation demand significant computational resources
- Integrating multimodal sensor, interaction, and system data remains complex
- VR environments may increase cognitive workload and learning difficulty for novice users.

4 VRDT Implementation for CPMS



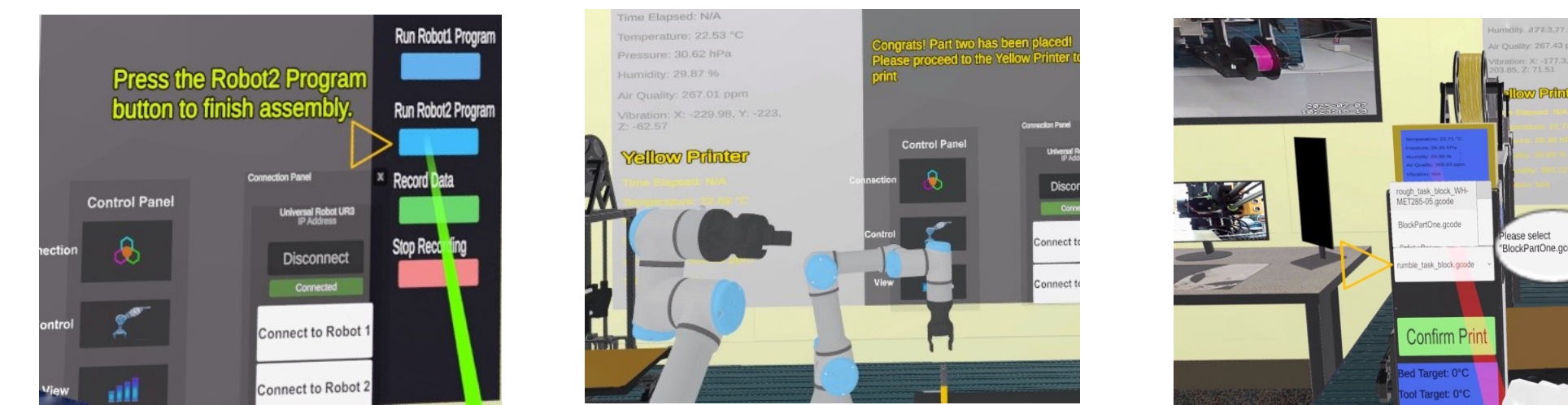
- VRDT integrates 3D printers, collaborative robots, sensors, and conveyors within an immersive virtual environment.
- Unity-based digital twin enables real-time monitoring of printer status, location, and temperature.
- ZeroMQ framework supports real-time bi-directional data transfer between physical and virtual systems.
- Users can remotely control manufacturing operations and visualize live production processes in VR

5 VR 3D Printing Digital Twin



- The physical twin consists of collaborative robots, 3D printers, conveyors, sensors, and real-time monitoring devices
- The digital twin replicates the manufacturing system in Unity, enabling immersive visualization, interaction, and real-time system synchronization
- Tasks included robot control, part handling, and 3D printer operation in virtual and physical systems.

6 EXPERIMENT IN VR Digital Twin



Physiological Data

- Electrodermal activity
- Blood Vessel Pulse

Video Recordings

Interaction Log Data

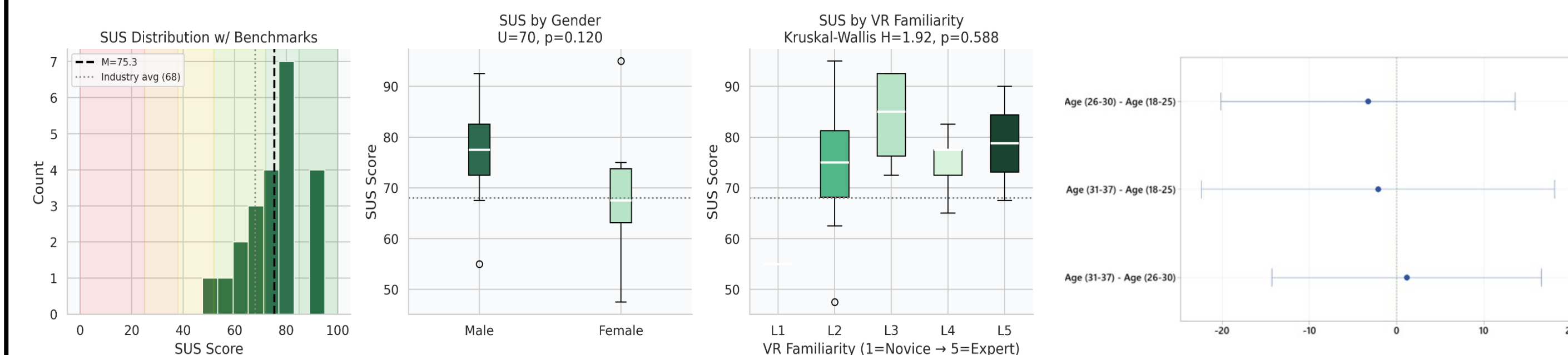
Survey Data

- Demographics
- SIM TLX
- System Usability Scale (SUS)

7 REAL-TIME DATA TRANSFER

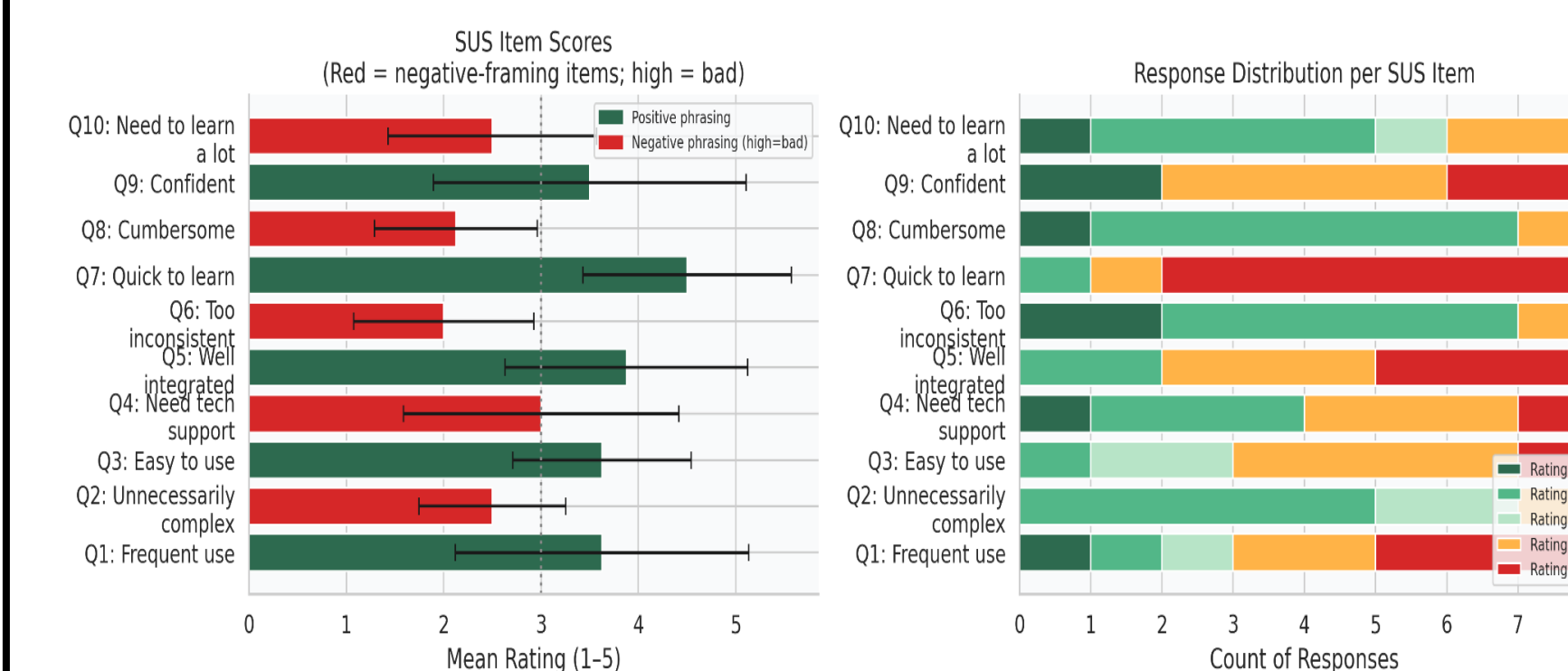
- Eye tracker** captures gaze fixation location, duration, pupil dilation, etc.
- Facial tracker** captures facial expressions such as smile, anger, fear, etc.
- System log data**
- Environment sensor** data (e.g., temperature, pressure, humidity, platform location, extruder movement, operator commands)
- Microphone** facilitates a multiplayer game environment

8 SUS Analysis



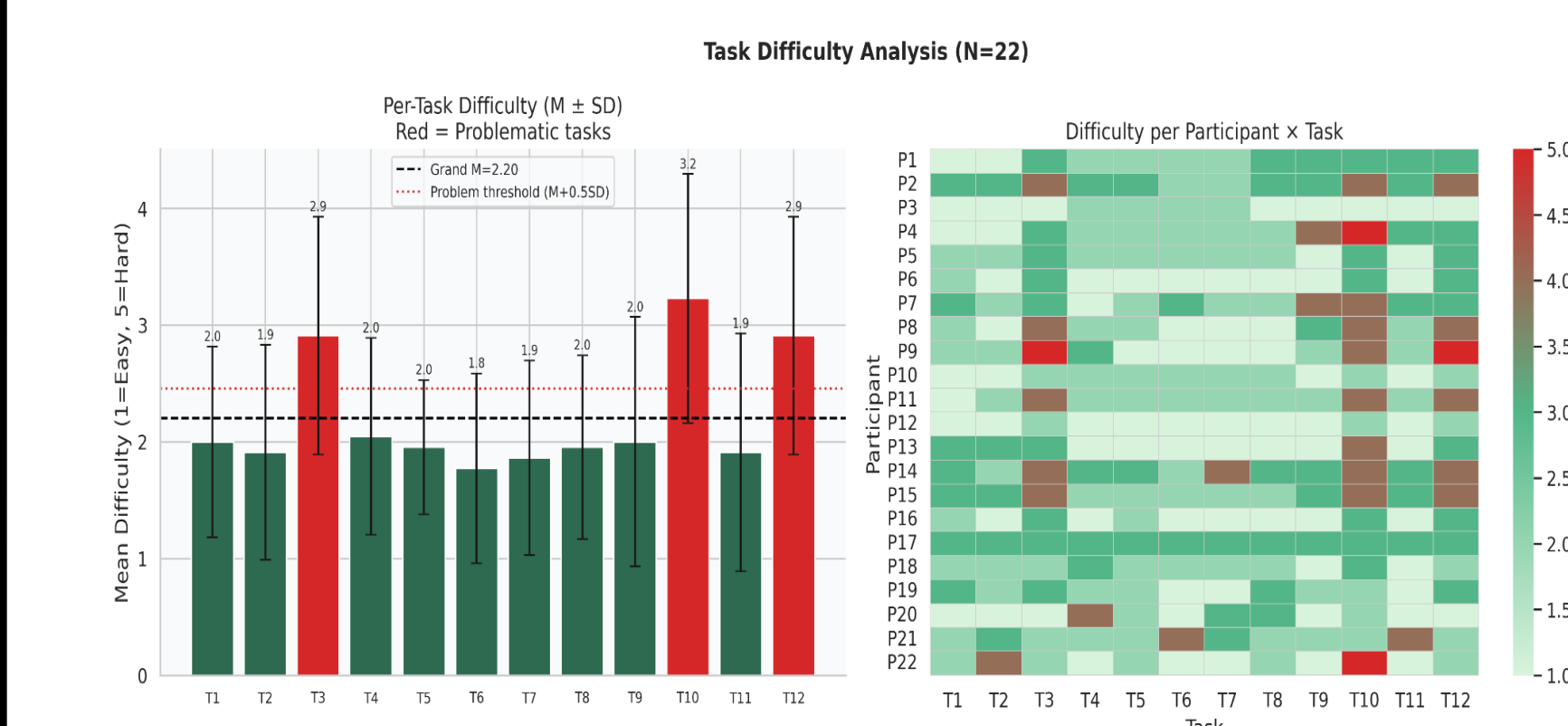
- Mean SUS score of 75.3 indicates good overall usability and positive user acceptance of the VRDT system.
- Users with higher VR familiarity reported better usability difficulty during interaction.
- No significant usability differences were observed across age groups

9 SUS Item Level Analysis



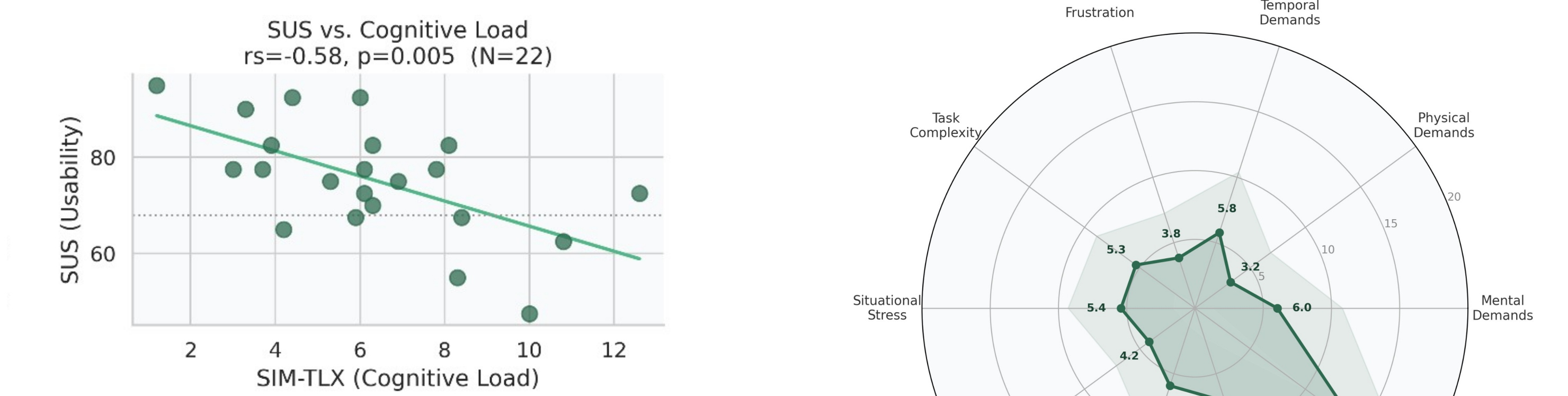
- Users found the VRDT reasonably easy to learn with well-integrated features.
- Some participants perceived interface complexity and needed additional guidance

10 Task Difficulty Analysis



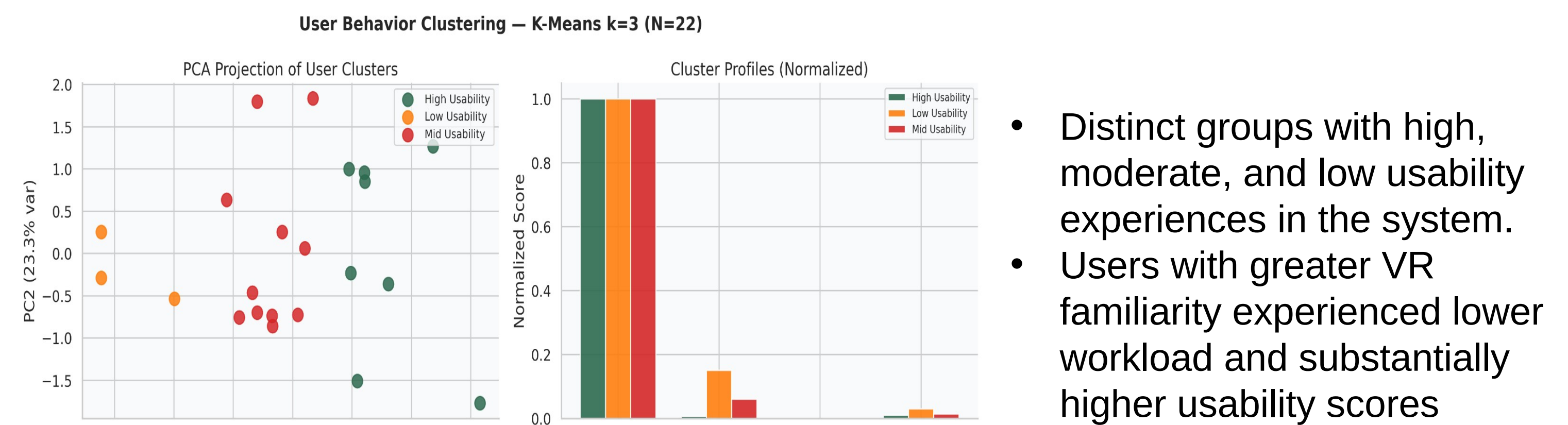
- Overall task difficulty remained moderate across the tasks.
- Difficulty patterns for Tasks 3, 10 and 12 were consistent across participants, indicating potential interface-level usability issues

11 SIM-TLX Analysis and SUS



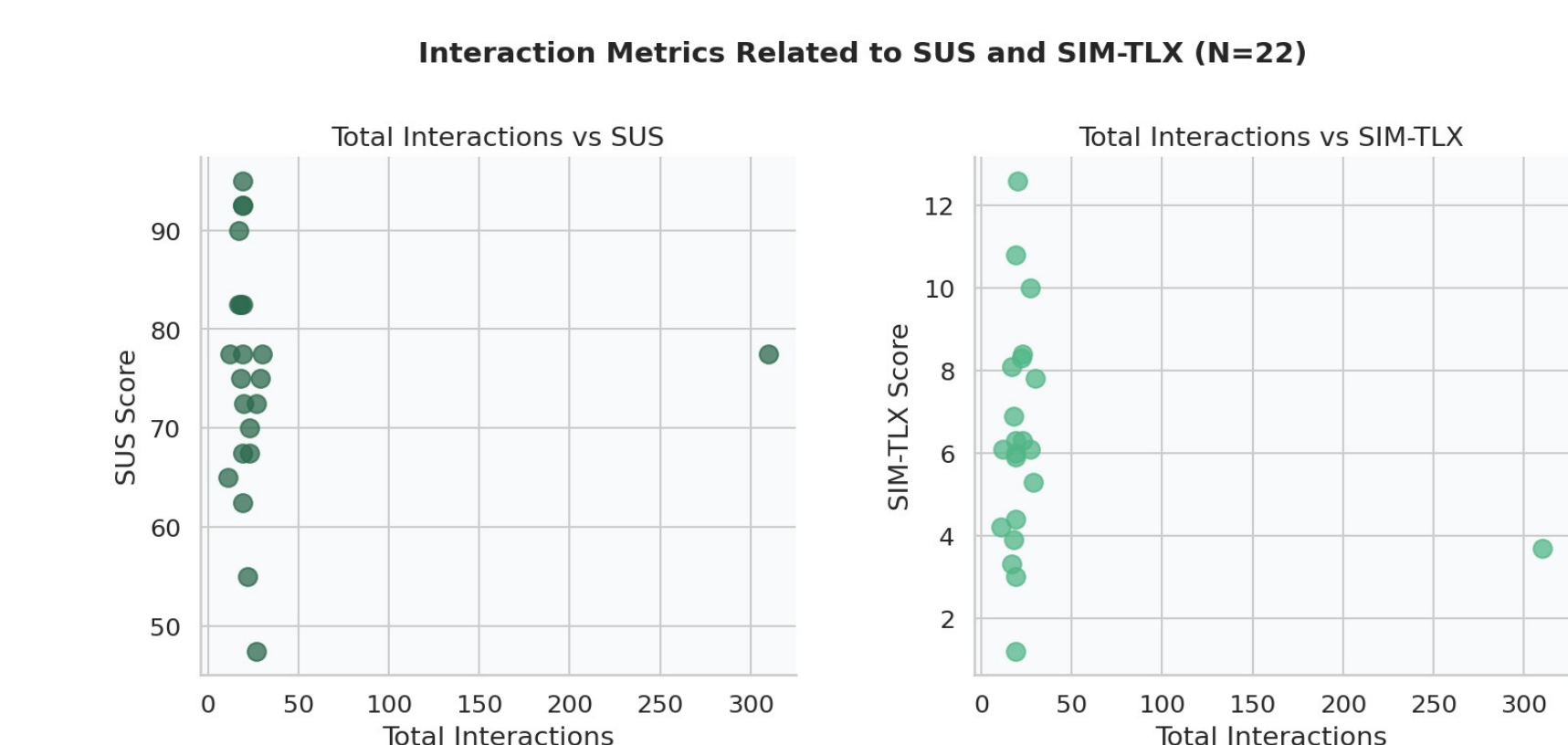
- Higher perceived workload was associated with lower usability.
- Strong immersion and presence, while users reported lower task control during interaction.

12 User Behavior Clustering



- Distinct groups with high, moderate, and low usability experiences in the system.
- Users with greater VR familiarity experienced lower workload and substantially higher usability scores

13 Interaction Log and SUS/SIM-TLX Relationship



- Little relationship between interaction frequency and perceived usability or cognitive workload scores.
- Excessive repeated interactions by some users indicated potential interface confusion and feedback limitations.

14 CONCLUSIONS

- The VRDT system demonstrated good usability and immersive user experience for manufacturing interaction tasks.
- Cognitive workload and VR familiarity significantly influenced performance and usability

Limitations include:

- Limited sample size and diversity.
- Potential simulation sickness.
- Time-intensive VRDT development and debugging process

Future work should focus on:

- Enhancing system usability
- Integrating data management and analytics
- Advancing adaptive VR system design

15 ACKNOWLEDGEMENT

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Dr. Rumena Begum



Dr. Faisal Aqlan

Siddhi Kabadi